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Application Note

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lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards

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Abstract

Motivation: The log-rank test is the standard for comparing two survival distributions but is optimally powered only under proportional hazards (PH). Modern biological settings — immune checkpoint immunotherapy, aging-genetics models, and hormone-responsive pharmacology — increasingly produce temporally concentrated hazard departures. Existing weighted log-rank alternatives (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington) pre-specify a temporal emphasis, rendering them insensitive to departures outside their focal window.
Results: We introduce `lifelines-hc`, a Python package extending the `lifelines` survival library with the Higher Criticism (HC) test for right-censored two-sample data. HC evaluates the collective extremity of per-interval hypergeometric p-values and detects any localised hazard departure without committing to a temporal weight. Across three biological case studies — an immuno-oncology trial (CheckMate 057 PFS, $n = 582$), a simulated *Drosophila* late-life quantitative trait locus ($n = 1600$), and an adjuvant bisphosphonate trial (AZURE DFS, $n = 3359$) — HC achieves $p \leq 0.013$ while the log-rank test and all four weighted alternatives return $p \geq 0.10$.
Availability and Implementation: `lifelines-hc` is freely available under the MIT licence at `https://github.com/TODO/lifelines-hc` and via `pip install lifelines-hc`; it requires Python ≥ 3.8 and `lifelines` ≥ 0.29. Full analysis scripts and per-method results tables are provided as Supplementary Data.

Key words: survival analysis; higher criticism; non-proportional hazards; hypothesis testing; Python; immuno-oncology

Introduction

The Kaplan-Meier estimator [?] and the log-rank test [?] remain the workhorses of two-sample survival analysis in biomedical research. The log-rank statistic sums weighted observed-minus-expected event counts across all failure times and is asymptotically optimal when the hazard ratio between groups is constant throughout follow-up — the proportional hazards (PH) assumption [?].

Modern biological and clinical data increasingly violate PH in structured ways. Immune checkpoint inhibitors require an immune-priming phase before generating durable tumour suppression, producing survival curves that cross before diverging. This pattern is characteristic of the second-line NSCLC trial CheckMate 057, in which nivolumab showed an early progression excess relative to docetaxel before the immunotherapy arm gained a durable advantage [?]. In aging genetics, mutation-accumulation theory [?] predicts alleles whose deleterious effects are expressed exclusively in post-reproductive life, creating narrow late-life mortality hot-spots that global averaging dilutes. In pharmacology, benefits conditional on a patient subgroup generate composite hazard differences that partially cancel: in the AZURE trial of adjuvant zoledronic acid, a bisphosphonate, the drug's bone-microenvironment benefit requires the low-oestrogen state of postmenopause and was absent in the many premenopausal patients enrolled [?].

Weighted log-rank tests were developed to address non-PH alternatives. Gehan-Wilcoxon [?] and Peto-Prentice [?] down-weight late events to increase sensitivity to early hazard differences; Tarone-Ware [?] applies an intermediate weight; Fleming-Harrington [?] provides a configurable weight via parameters (p, q). However, each test concentrates power on a specific temporal region, so a test optimised for early differences is correspondingly insensitive to late ones, and no single pre-specified weight simultaneously covers all non-PH patterns.

Higher Criticism (HC; **?**) is a meta-test that detects non-random concentrations of small p-values within a collection of tests. Applied to survival analysis [?], HC divides the follow-up into equal-width intervals, derives a per-interval hypergeometric p-value comparing event rates between the two groups, and assesses whether the most deviant subset of these p-values is collectively implausible under the global null. Because HC does not pre-weight any particular time window it retains power for concentrated hazard departures wherever they occur along the follow-up axis.

Despite this theoretical advantage, an accessible Python implementation integrated with standard survival analysis workflows has been lacking. We present `lifelines-hc`, which fills this gap by extending the widely-used `lifelines` library [?].

The `lifelines-hc` package

Higher Criticism test for survival data

Given two groups with right-censored observations (T_i, E_i, G_i), partition the follow-up range into K equal-width intervals. For interval k, let n_k be the total number of events, r_k^A and r_k^B the subjects at risk in groups A and B, and x_k the number of events in group B. Under the null of equal group-specific hazards,

$$x_k \;\Big|\; n_k,\, r_k^A,\, r_k^B \;\sim\; \text{Hypergeometric}\Big(r_k^A + r_k^B,\; r_k^B,\; n_k\Big), \quad (1)$$

yielding a one-sided interval p-value p_k. Ordering the K p-values as $p_{(1)} \le \cdots \le p_{(K)}$, the HC statistic is

$$\mathrm{HC}_K \;=\; \max_{1\le j\le \lfloor \gamma K\rfloor} \frac{j/K - p_{(j)}}{\sqrt{p_{(j)}\left(1-p_{(j)}\right)/K}}, \quad (2)$$

where $\gamma \in (0,1)$ (default 0.2) restricts scanning to the most extreme fraction of interval p-values [?]. A global p-value is obtained by permuting group labels ($n_{\text{perms}} = 500$ by default). For the two-sided alternative, the minimum of the two one-sided interval p-values replaces p_k in (2).

Software design and API

`lifelines-hc` exposes a function-based API consistent with `lifelines.statistics`, accepting NumPy arrays of event times and indicators:

```
from lifelines_hc import (higher_criticism_test,
                          fisher_combination_test
                          ,
                          min_p_test,
                          suspected_deviations)

result = higher_criticism_test(
    T_A, T_B,
    event_observed_A=E_A, event_observed_B=E_B,
    n_intervals_to_pool=100, n_permutations=500,
    alternative='both', gamma=0.2, seed=42)

print(result.p_value)        # permutation-
    calibrated p-value
print(result.test_statistic)  # HC statistic
    value
```

The companion `suspected_deviations()` function returns a `pandas.DataFrame` of per-interval p-values with a `suspected` boolean column marking HC-flagged intervals, enabling overlay of diagnostic shading on Kaplan-Meier plots. Two complementary omnibus tests are also provided: Fisher combination [?], which aggregates evidence via $-2\sum_k \log p_k$, and a Bonferroni-corrected MinP test. All three share the same permutation infrastructure. The package additionally ships with plotting utilities (`plot_km_with_hc`, `plot_pvalue_profile`) and a convenience wrapper `run_all_tests()` that benchmarks HC, Fisher, MinP, log-rank, and the four weighted alternatives in a single call, returning a `pandas.DataFrame` for easy comparison.

Results

We demonstrate `lifelines-hc` on three biological datasets with distinct non-PH structures (Figure 1), benchmarking HC against the log-rank test and four weighted alternatives. Full per-method p-value tables and the *Drosophila* case study are provided in Supplementary Data.

Immuno-oncology: crossing survival curves

CheckMate 057 [?] randomised 582 patients with advanced non-squamous NSCLC to nivolumab ($n = 292$) or docetaxel ($n = 290$). Progression-free survival showed crossing curves: docetaxel provided a short-term advantage during immune priming, subsequently overtaken by the durable nivolumab response (Figure 1A). Individual patient data were reconstructed from the published Kaplan-Meier curve via the algorithm of **?** as implemented in the `kmdata` R package.

The log-rank test is non-significant ($p = 0.351$) as the early excess and late benefit partially cancel. Among the weighted alternatives, Gehan-Wilcoxon ($p = 0.041$) and Peto-Prentice ($p = 0.049$) reach borderline significance by detecting the early crossing, but Tarone-Ware ($p = 0.358$) and Fleming-Harrington (1,1) ($p = 0.256$) do not. HC achieves $p = 0.002$ and characterises the full crossing structure with far greater power (Figure 1B) by flagging both the early divergence and late separation windows independently.

Aging genetics: a narrow late-life mortality hot-spot

To represent late-life QTL biology in *Drosophila melanogaster* [**??**], we generated a synthetic piecewise-exponential dataset ($n = 800$ per genotype) with baseline median lifespan 60 days, 90-day censoring, and a 4× elevated hazard exclusively in days 60–64 (the QTL window), with identical hazards outside this window. With 75 flanking null intervals, the log-rank test is non-significant ($p = 0.104$). All four weighted alternatives also fail: early-weighting tests (Gehan-Wilcoxon and Peto-Prentice, both $p = 0.203$) are insensitive to the late signal, and Fleming-Harrington (1,1), which emphasises mid-follow-up, reaches only $p = 0.060$. HC ($p = 0.002$) identifies the concentrated spike in the interval p-value profile (Supplementary Figure S1), demonstrating that its scanning approach is not penalised by the long null early period.

Adjuvant bisphosphonate therapy: a menopause-dependent effect

The AZURE trial [**??**] randomised 3359 early-stage breast cancer patients to standard adjuvant therapy with ($n = 1681$) or without ($n = 1678$) zoledronic acid. Disease-free survival showed no significant log-rank difference ($p = 0.305$) because the drug's bone-microenvironment benefit depends on the low-oestrogen state of postmenopause, which was absent in the many premenopausal patients enrolled at baseline. As these patients progressively transitioned to postmenopause during the decade-long follow-up, the hazard difference accumulated in a specific mid-to-late window (months 20–60) rather than uniformly (Figure 1C–D).

All four weighted alternatives are non-significant ($p \ge 0.28$): no single temporal weight captures a pattern distributed across this window while remaining unaffected by the null early period. HC ($p = 0.012$) detects this scattered but collectively significant signal. The result is validated by published subgroup analyses showing significant DFS benefit in postmenopausal patients [**??**], confirming that HC is detecting a genuine biological effect.

Across the three case studies, the two borderline significances for Gehan-Wilcoxon and Peto-Prentice in CheckMate 057 are the only instances where a weighted alternative reaches $p < 0.05$, and these are attributable to those tests' early emphasis coincidentally aligning with the crossing-curve signal rather than to any general superiority of the weighted approach. Fisher combination ($p \leq 0.04$ in all three domains) and MinP ($p \leq 0.022$ in two of three) are also significant, consistent with the sparse, localised nature of the underlying signals.

Conclusion

`lifelines-hc` provides a general-purpose HC test for two-sample right-censored survival data within the Python ecosystem. Unlike weighted log-rank alternatives, HC requires no a priori specification of the temporal window in which a hazard departure is expected and retains power for concentrated departures wherever they occur. The package integrates transparently with `lifelines` workflows, returns diagnostic interval-level information identifying which time windows drive the test result, and includes Fisher combination and MinP tests as complementary omnibus alternatives. Together, these tools offer a practical complement to standard survival testing in any setting where the proportional hazards assumption is uncertain.

Acknowledgements

TODO.

Conflict of Interest

None declared.

Funding

TODO.

Data availability

Individual patient data for CheckMate 057 and the AZURE trial were reconstructed from published Kaplan-Meier curves using the algorithm of **?** as implemented in the `kmdata` R package (`https://github.com/raredd/kmdata`). The synthetic *Drosophila* dataset is generated by the reproducible analysis scripts distributed with the package. All analysis code is available at `https://github.com/TODO/lifelines-hc/tree/main/AppNote`.

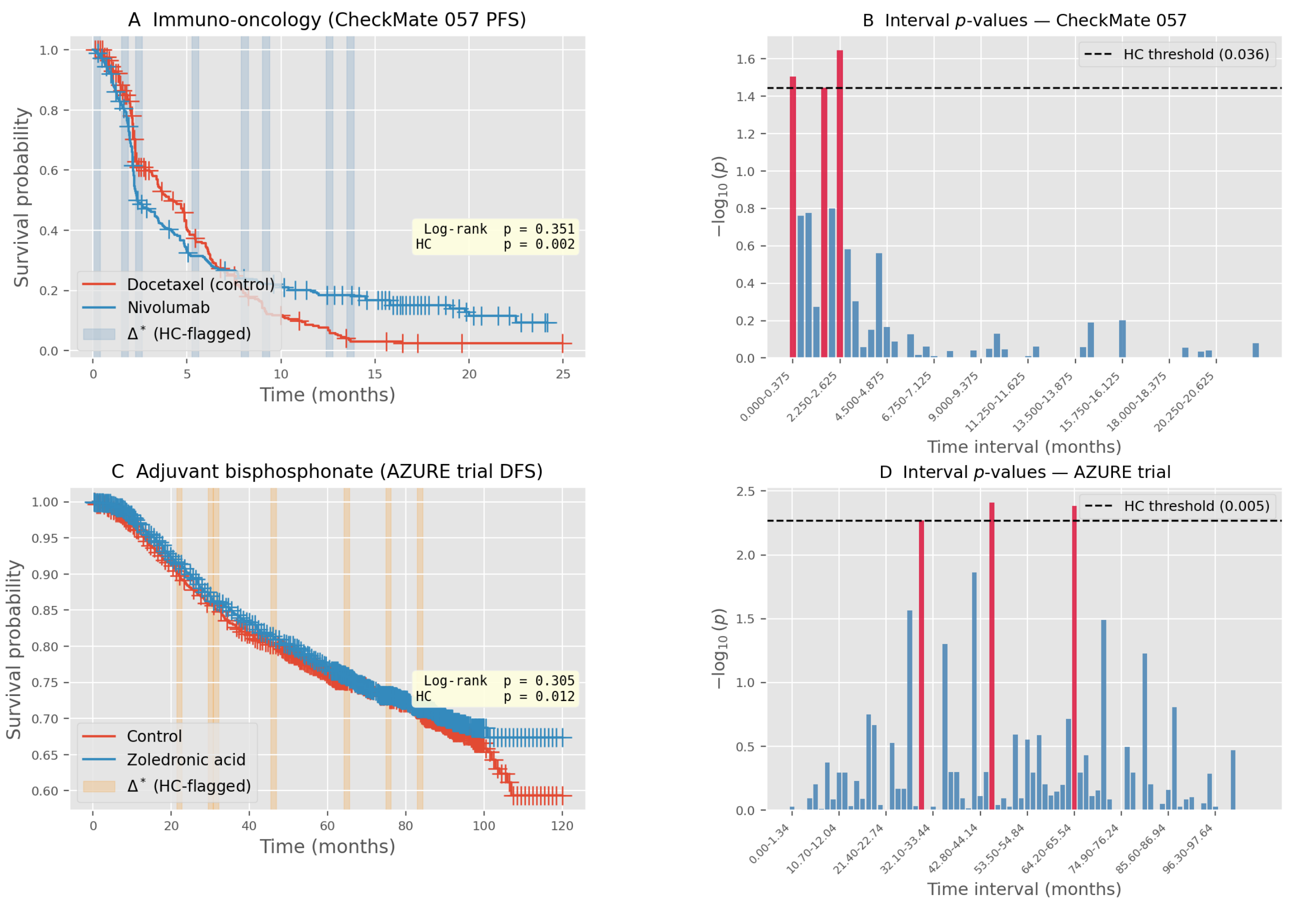

Fig. 1. Higher Criticism detects non-proportional hazard departures across two biological domains. (A) Kaplan-Meier progression-free survival curves for CheckMate 057 (nivolumab vs. docetaxel, $n = 582$) with HC-flagged intervals shaded in blue; log-rank $p = 0.351$, HC $p = 0.002$. Curves cross at $\sim$3 months before diverging. **(B)** Per-interval $-\log_{10}$(hypergeometric p-value) for CheckMate 057. Bars exceeding the HC threshold (dashed, permutation- calibrated) are highlighted in red, marking the early-crossing and late-separation windows. **(C)** Kaplan-Meier disease-free survival for the AZURE trial (zoledronic acid vs. control, $n = 3359$) with HC-flagged intervals in orange; log-rank $p = 0.305$, HC $p = 0.012$. The signal reflects a menopause-dependent benefit accumulating in months 20–60. **(D)** Per-interval $-\log_{10}$(hypergeometric p-value) for AZURE. The signal is distributed across the mid-follow-up range, outside the reach of any single-weight log-rank alternative. Individual patient data reconstructed via **?**.